LOVELY PROFESSIONAL UNIVERSITY

Sample Question Set - Unit 5

Course Code: CSE252	Unit: 5
Course Title: INTRODUCTION TO ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING	Unit Title: Model Evaluation and Improvement
Course Outcome: CO5	Unit Mapped RBT: L3 (Apply)
Paper: Sample Subjective Questions	Total Questions: 10
Maximum Marks: 10 x 10 = 100	Suggested Duration: Not timed - practice set

Subjective Questions (10 x 10 = 100 Marks)

Q1. Apply the confusion matrix and the classification metrics to a screening problem of your own choosing. (10 Marks)

(a) State the four entries of the matrix and how accuracy, precision and recall are computed from them.

(b) Use a worked set of counts to compute the three, and say what each one tells you about the model.

[L3 - Apply]

Q2. Compare precision with recall and explain the exchange between them. (10 Marks)

(a) State what each of the two measures and over what set of records.

(b) Use the decision threshold to explain how raising one lowers the other, and when each should be favoured.

[L3 - Apply]

Q3. Examine a classifier reporting a very high accuracy on a dataset in which one class is rare. (10 Marks)

(a) State how so high a figure can be obtained without the model being useful.

(b) Use two other measures to describe how you would expose the problem and report it honestly.

[L3 - Apply]

Q4. Use cross validation to obtain a dependable estimate of a model's performance. (10 Marks)

(a) State how k-fold cross validation proceeds and what is reported at the end.

(b) Use a small dataset to explain why this is preferred to a single train-test split.

[L3 - Apply]

Q5. Apply the regression metrics to a result and interpret what they say together. (10 Marks)

(a) State what mean absolute error, root mean squared error and R squared each measure.

(b) Use a result with a small mean absolute error and a large root mean squared error to describe what has happened.

[L3 - Apply]

Q6. Use hyperparameter tuning to improve a model, and describe the procedure end to end. (10 Marks)

(a) State the difference between a parameter and a hyperparameter, with an example of each.

(b) Use a search strategy and a validation scheme to describe how the settings would be chosen and then confirmed.

[L3 - Apply]

Q7. Examine a model whose cross-validated score is far above the performance it shows in service. (10 Marks)

(a) State two distinct explanations for the gap and how each of the two arises.

(b) Use one of the two to describe the checks you would run and the correction you would make.

[L3 - Apply]

Q8. Apply the ideas of Explainable Artificial Intelligence to a system that decides applications. (10 Marks)

(a) State the difference between a global and a local explanation, and who needs each of the two.

(b) Use a rejected application to describe what the organisation should be able to show, and why.

[L3 - Apply]

Q9. Use the decision threshold of a classifier to trade recall against precision. (10 Marks)

(a) State what the threshold does and what happens to each metric as it is raised and lowered.

(b) Use a fraud screening system to justify where you would place the threshold, and what you would monitor.

[L3 - Apply]

Q10. Relate the choice of evaluation metric to the cost of the two kinds of error in a given application. (10 Marks)

(a) State what a false positive and a false negative would cost in a named application.

(b) Use those costs to select a metric to optimise, and explain why accuracy would be the wrong choice.

[L3 - Apply]

----- End of Sample Question Set -----